

Nikita Kazeev

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Research Scientist at the intersection of AI and Physics

WORK EXPERIENCE

PERFOMAX *AI Advisor* 2026–present

NATIONAL UNIVERSITY OF SINGAPORE *Postdoc under [Kostya Novoselov](#)* 2022–present

Leadership

- Program Chair for the [AI4X 2025/6 conferences](#)
- Main organiser of the [ICLR 2025 Workshop on Machine Learning Multiscale Processes](#)
- Co-PI of a US\$3.4m AI Singapore grant on multiscale machine learning

Wyckoff Transformer, a generative model for materials [ICML 2025](#) · [GitHub](#)

- Developed a coordinate-free permutation-invariant autoregressive Transformer encoder model for material generation and property prediction based on symmetry inductive bias
- Increased materials discovery yield (S.U.N.) by 1.24x
- Cut inference to 0.05 GPU-ms/structure — 4 orders of magnitude faster than prior models
- Implemented the model in PyTorch; production-ready package published on [PyPI](#)
- Orchestrated large-scale (10k structures) DFT computations with VASP on an HPC cluster
- Led a team of 6

(CERN | YANDEX → HSE UNIVERSITY) *Researcher* 2014–2022

[2025 Breakthrough Prize in Fundamental Physics](#) as a member of LHCb.

Generative models uncertainty estimation [paper](#) · [conference](#) · [code 1](#) · [code 2](#)

- Developed the first methods for estimating uncertainty of conditional GANs
- Led a team of 3 students

Generative models for fast simulation [paper 1](#) · [paper 2](#) · [paper 3](#) · [code](#)

- Built a GAN for high-fidelity, tabular-data simulation of a Cherenkov detector
- Reached $10^5\times$ speed-up over the full Geant4 simulation

EDUCATION

HSE University & Sapienza Università di Roma <i>PhD in Computer Science and Physics (Double Degree)</i> Supervisors: Andrey Ustyuzhanin & Barbara Sciascia Thesis: Machine Learning for particle identification in the LHCb detector	2016–2020
Product Management course at Yandex <i>Focused coursework in product strategy and execution.</i>	2018
Yandex School of Data Analysis <i>Master's-level CS programme</i> Algorithms, machine learning, deep learning, and distributed systems.	2013–2015
Moscow Institute of Physics and Technology <i>MS in Physics</i> Optimisation of data processing of the LHCb experiment.	2014–2016
<i>BS in Physics</i> Wave-packet molecular dynamics of electrons in nonideal plasma.	2010–2014
International Junior Science Olympiad (IJSO) Silver medal	Korea, 2008

MENTORSHIP

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- Mentored 9 students ([1](#), [2](#), [3](#), [4](#), [5](#), [6](#), [7](#), [8](#), [9](#)), 3 interns, and student workshop projects ([1](#), [2](#)).

TEACHING & OUTREACH

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- Pitch research to the general public through major newspaper coverage ([1](#), [2](#)), blog posts ([1](#), [2](#)), a [science slam](#), interviews ([1](#), [2](#)), and a public [talk](#).
 - Lecturer in the [award-winning](#) Coursera Advanced Machine Learning [specialisation](#)
 - Delivered more than 20 [conference talks](#).

SERVICE

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- Reviewer for NeurIPS, RSC Advances, Machine Learning: Science and Technology, Digital Discovery
 - Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

SKILLS

- **Generative modelling for scientific data** – GANs, autoregressive transformers, and diffusion; from fast detector simulation to symmetry-aware crystal generation.
- **Uncertainty quantification** – built the first methods for estimating the uncertainty of conditional GANs; calibrated generative and discriminative models.
- **ML on structured, non-text data** – tabular, graph, and atomistic representations, spanning gradient boosting to transformers.
- **Research leadership** – lead multidisciplinary teams, chair conferences, and mentor students from idea to publication.

ML & AI • Python • PyTorch • HPC • Physics • Research Writing & Speaking • Leadership & Mentorship

PRESENTING AT ICML 2026

July 11, 12:15–13:30 – Nikita Kazeev and Ian Babich. [“Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments”](#). In: *Forecasting as a New Frontier of Intelligence Workshop at ICML 2026*.

July 11, 15:00–16:30 – Nikita Kazeev and Phan Bui Nhat Huyen. [“Position: Align AI to Our Aspirations, Not Our Flaws”](#). In: *Pluralistic Alignment Workshop at ICML 2026*.

July 11, 15:30–17:00 – Nikita Kazeev and Andrey Ustyuzhanin. [“The Geometry of Reasoning Failure: Predicting Agent Errors from Semantic Scale Trajectories”](#). In: *Statistical Frameworks for Uncertainty in Agentic Systems Workshop at ICML 2026*.

July 10, 11:40–13:30 & 16:10–17:00 – Nikita Kazeev and Andrey Ustyuzhanin. [“Uncertainty Quantification for LLM Agents via Semantic Abstraction Trajectories”](#). In: *Structured Probabilistic Inference & Generative Modeling Workshop at ICML 2026*.

SELECTED PUBLICATIONS

Nikita Kazeev, Wei Nong, Ignat Romanov, Ruiming Zhu, Andrey E Ustyuzhanin, Shuya Yamazaki, and Kedar Hipalgaoonkar. [“Wyckoff Transformer: Generation of Symmetric Crystals”](#). In: *Proceedings of the 42nd International Conference on Machine Learning*. Ed. by Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu. Vol. 267. Proceedings of Machine Learning Research. PMLR, 13–19 Jul 2025, pp. 29495–29526.

Denis Derkach, Nikita Kazeev, Fedor Ratnikov, Andrey Ustyuzhanin, and Alexandra Volokhova. [“Cherenkov detectors fast simulation using neural networks”](#). In: *Nuclear Instruments and Methods in Physics Research Section A* (2019). Alphabetic order, I’m the corresponding author.